

Button Password Lock System Using Arduino Uno

Project Aim

To design and implement a Button Password Lock System using an Arduino Uno, three push buttons, a buzzer, a green LED, a red LED, and a 16x2 LCD display. The system authenticates a user based on a predefined button sequence and provides visual and audio feedback for successful and unsuccessful attempts.

Project Overview

The project uses three push buttons to enter a password sequence. The predefined password is Button 2 → Button 1 → Button 3. After three button presses, the Arduino verifies the sequence. If the sequence is correct, the system displays ACCESS GRANTED on the LCD and Serial Monitor, turns on the green LED, and plays a success tone. If the sequence is incorrect, it displays ACCESS DENIED, turns on the red LED, and plays an error tone. The system automatically resets for the next user.

Components Used

- Arduino Uno
- 3 Push Buttons
- 16x2 LCD Display
- 10kΩ Potentiometer
- Green LED
- Red LED
- Piezo Buzzer
- 220Ω Resistors
- Breadboard and Jumper Wires

Construction and Wiring Details

Buttons: Button 1 → D2, Button 2 → D3, Button 3 → D4 using INPUT PULLUP configuration.

Green LED → D8 through 220Ω resistor.

Red LED → D9 through 220Ω resistor.

Buzzer → D10.

LCD: RS→D12, E→D11, DB4→D5, DB5→D6, DB6→D7, DB7→D13, RW→GND, VCC→5V, GND→GND.

Potentiometer: Ends to 5V and GND, middle pin to LCD V0 for contrast control.

Code Structure / Program Flow

1. Initialize LCD, LEDs, buzzer, and button inputs.
2. Display ENTER PASSWORD on LCD.
3. Detect button presses and store the entered sequence.
4. After three inputs, compare with the predefined password.
5. If correct: display ACCESS GRANTED, activate green LED and success tone.
6. If incorrect: display ACCESS DENIED, activate red LED and error tone.
7. Reset the system and wait for the next user.

Source Code

```
LiquidCrystal lcd(12, 11, 5, 6, 7, 13);

const int button1 = 2;

const int button2 = 3;

const int button3 = 4;

const int greenLED = 8;

const int redLED = 9;

const int buzzer = 10;

int password[3] = {2, 1, 3};

int entered[3];

int count = 0;

void setup()
{
    pinMode(button1, INPUT_PULLUP);
    pinMode(button2, INPUT_PULLUP);
    pinMode(button3, INPUT_PULLUP);
    pinMode(greenLED, OUTPUT);
    pinMode(redLED, OUTPUT);
    pinMode(buzzer, OUTPUT);
    Serial.begin(9600);
    lcd.begin(16, 2);
    lcd.print("ENTER");
    lcd.setCursor(0, 1);
    lcd.print("PASSWORD");
}

void loop()
{
    if (digitalRead(button1) == LOW)
    {
        entered[count++] = 1;
        delay(250);
    }
}
```

```

if (digitalRead(button2) == LOW)
{
    entered[count++] = 2;
    delay(250);
}
if (digitalRead(button3) == LOW)
{
    entered[count++] = 3;
    delay(250);
}
if (count == 3)
{
    checkPassword();
}
}

void checkPassword()
{
    bool correct = true;
    for (int i = 0; i < 3; i++)
    {
        if (entered[i] != password[i])
        {
            correct = false;
            break;
        }
    }
}

lcd.clear();

if (correct)
{
    Serial.println("ACCESS GRANTED");
}

```

```

    lcd.print("ACCESS");

    lcd.setCursor(0, 1);

    lcd.print("GRANTED");
digitalWrite(greenLED, HIGH);
tone(buzzer, 1000, 500);
delay(2000);
}

else
{
    Serial.println("ACCESS DENIED");
    lcd.print("ACCESS");

    lcd.setCursor(0, 1);

    lcd.print("DENIED");
digitalWrite(redLED, HIGH);
tone(buzzer, 300, 1000);
delay(2000);
}

resetSystem();
}

void resetSystem()
{
    digitalWrite(greenLED, LOW);

    digitalWrite(redLED, LOW);

    count = 0;
    for (int i = 0; i < 3; i++)
    {
        entered[i] = 0;
    }

    lcd.clear();

    lcd.print("ENTER");

    lcd.setCursor(0, 1);

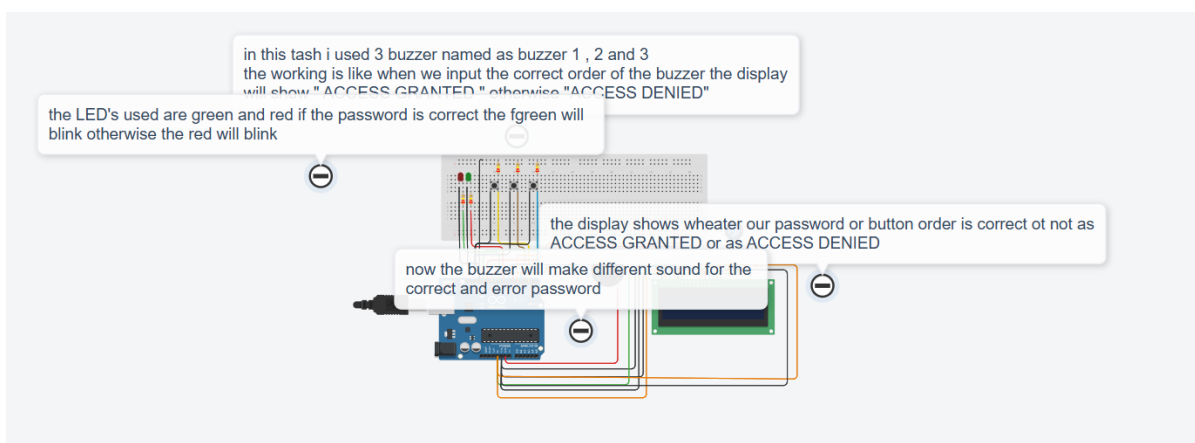
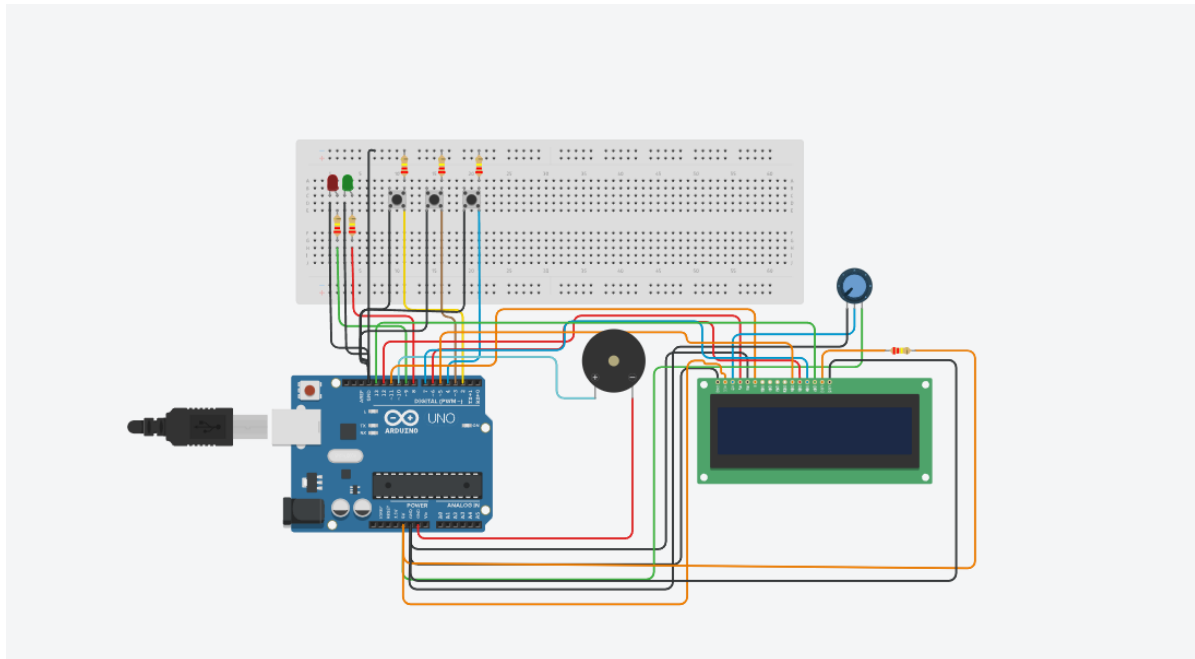
    lcd.print("PASSWORD");

```

```
Serial.println("READY FOR NEXT USER");
}
```

Project Photograph(s)

Insert screenshots or photographs of the completed Tinkercad circuit, wiring layout, LCD display output, and simulation results in this section.



The links

Google drive : https://drive.google.com/drive/folders/1xD0VgLU7slZdNRq-6pnJe22Dulm-rrJw?usp=drive_link

Github repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>

Tinkercad simulation :

https://www.tinkercad.com/things/hNTULseij2i/editel?returnTo=%2Fdashboard&sharecode=M_3S3WYhUWsa-6r2s4jZ8Ue_1-i7--iYvAW1UJe8shY